

Experimental Design

Genomics best-practices reference PDF

Experimental Design

RNA-Seq

- Power Calculation prior to experiment
- Paired-end sequencing always recommended
- Low tumor purity requires more reads

ChIP-Seq

- Power Calculation prior to experiment
- Input depth should be twice as high as ChIP depth

Power Calculation prior to experiment

- ATAC-Seq
- Fragment length < 100 = Nucleosome free
- Fragment length ~ 150 -200 = Mono-nucleosome bound
- Fragment length ≥ 300 = Di/Multi-nucleosome bound

1 Variable

- Quantitative
- Categorical
- Parametric
- Non-Parametric
- > 2 groups
- 2 groups
- One-Sample Wilcoxon Test
- Example: TP53 expression against known expected expression (skewed distribution)
- One-Sample Student's T-test
- Example: TP53 expression against known expected expression
- One-Proportion Test
- Example: Gene mutation proportion matches expected
- Chi-Squared Goodness of Fit

1 Variable

- Example: Codon Frequencies against expected frequencies

>2 Variables

- Quantitative
- Categorical
- 2 groups
- > 2 groups
- Binary Logistic Regression
- Example: Predicting tumor presence based on gene expression
- Multiple Linear Regression
- Example: Prediction tumor size based on expression of 4 genes
- Multinomial Logistic Regression
- Example: Predicting tumor stage based on gene expression

2 Quantitative

- Non-Parametric
- Parametric
- Spearman Correlation
- Example: Co-expression with non-linear relationship
- Pearson Correlation
- Example: Co-expression with linear relationship

2 Qualitative

- 2 groups
- > 2 groups
- Independent
- Paired
- Expected frequencies < 5
- Expected frequencies < 5
- > 2 Paired Samples
- 2 Paired Samples
- Cochran's Q-Test
- Example: Mutation calling differ between >2 different variant callers
- McNemar's Test
- Example: Mutation calling differ between 2 different variant callers
- Fisher's Exact Test

2 Qualitative

- Example: Mutation presence associated with drug response? ($n < 5$)
- Chi-Square Test of Independence
- Example: Mutation presence associated with drug response? ($n \geq 5$)

1 Quantitative

- 1 Qualitative
- 2 Groups
- Independent
- Paired
- Parametric
- Nonparametric
- Parametric
- Nonparametric
- Unequal population variances
- Equal population variances
- Wilcoxon-Signed rank test
- Example: TP53 expression in tumor vs adjacent normal samples
- Paired Student's t-test

1 Quantitative

- Example: TP53 expression in tumor vs adjacent normal samples
- Student's t-test
- Example: TP53 expression in tumor vs normal samples
- Mann-Whitney U Test
- Example: TP53 expression in tumor vs normal samples (skewed distribution)
- Welch's t-test
- Example: TP53 expression in tumor vs normal samples

1 Quantitative 1 Qualitative

- >2 Groups
- Independent
- Paired
- Parametric
- Nonparametric
- Parametric
- Nonparametric
- Equal population variances
- Unequal population variances
- One-way ANOVA
- Example: TP53 expression across 3 conditions
- Repeated measures ANOVA
- Example: TP53 expression in metastatic vs tumor vs adjacent normal samples

1 Quantitative 1 Qualitative

- Kruskal-Wallis Test
- Example: TP53 expression across 3 conditions (skewed distribution)
- Welch's ANOVA
- Example: TP53 expression across 3 conditions (skewed distribution)
- Friedman Test
- Example: TP53 expression in metastatic vs tumor vs adjacent normal samples

References

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- Chromatin accessibility profiling by ATAC-seq <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9189070/>
- Statistics flow chart <https://statsandr.com/blog/what-statistical-test-should-i-do/>